Kopal Khare

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Summary

Computer Science undergraduate specializing in AI ML, focused on full-stack and backend engineering. Experienced in Java, Spring Boot, Next.js, React, TypeScript, REST APIs, PostgreSQL, Redis, and Docker. Built and deployed production-oriented applications involving scalable APIs, transactional systems, relational databases, and AI-powered workflows.

Technical Skills

Languages: Java, TypeScript, Python, SQL,
Backend & APIs: Spring Boot, REST APIs, FastAPI
Frontend: Next.js, React.js, Tailwind CSS
Data & Databases: PostgreSQL, Redis, Prisma ORM
AI & Data: OpenAI API, LLMs, RAG
Tools: Docker, Git, Postman, Flyway
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Projects

ClearLedger – Financial Ledger & Reconciliation Platform | Java, Spring Boot, PostgreSQL, Docker, Flyway, REST APIs

- Engineered a financial operations platform supporting account management, transactional transfers, reconciliation, and audit workflows with strong data integrity.
- Built settlement ingestion and deterministic two-way reconciliation workflows to identify transaction discrepancies and support structured resolution.
- Implemented trial-balance validation and historical account statements to provide reliable financial reporting and operational visibility.
- Maintained append-only audit history and validated core financial workflows with **49/49 automated tests**, ensuring consistent and traceable transaction processing.

Live Demo

AK Saarthi AI – AI-Powered Financial Advisory Platform | Next.js, React, TypeScript, PostgreSQL, Prisma, OpenAI API

- Built a full-stack financial advisory platform for managing client profiles, financial information, policies, and advisor workflows using Next.js and React.
- Designed relational PostgreSQL data models with Prisma ORM and integrated OpenAI APIs for contextual AI-assisted financial workflows.

Live Demo

Scalable URL Shortener & Analytics Platform | Java, Spring Boot, PostgreSQL, Redis, Docker

- Built a scalable URL management platform supporting custom aliases, link expiration, validation, and high-volume redirect handling.
- Designed analytics workflows to capture redirect activity and process traffic data asynchronously for usage insights.
- Implemented Redis caching, PostgreSQL indexing, and rate limiting to improve responsiveness and support reliable handling of high-volume traffic.

Archon AI – AI Engineering Intelligence Platform | Tree-sitter, Neo4j, RAG, pgvector, LangGraph

- Built an AI-powered platform that transforms software repositories into a queryable knowledge system for intelligent code understanding and engineering analysis.
- Implemented semantic retrieval and RAG workflows using a Neo4j knowledge graph to connect repository structure, dependencies, and source-code relationships.

Extra-Curricular

Blockchain Club June 2024 – Present
Core Member VIT Bhopal University

- Co-organized 4+ national-level technical events with 500+ participants in collaboration with Accelchain and Notion.
- Coordinated event planning, technical execution, and cross-functional logistics with student teams and external collaborators.

British Airways Data Science Job Simulation May 2026
Forage Virtual Experience Remote

- Analyzed customer review data using Python, performed exploratory analysis, and built a predictive model to identify actionable business insights.

Certifications

AWS Certified Solutions Architect Associate – AWS | IBM Machine Learning Specialization – Coursera
NPTEL Cloud Computing – IIT Kharagpur | NPTEL Internet of Things – IIT Kharagpur

Education

Vellore Institute of Technology July 2023 – July 2027
B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning) CGPA: 8.42/10